

A Special Advertising Supplement

March 31, 1997

THE YEAR 2000

SOLUTIONS SERIES

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DENIAL:

It's not a solution,

it's a gamble

Introduction

When the ghosts of the past come back to haunt us, it's a scary situation.

In order to save memory — once one of the most precious components of a computer system — programmers in the 1960s and 1970s represented the year in dates using two digits in code rather than four. This practice persisted into the 1980s and even the 1990s as firms tried to maintain backward compatibility with legacy systems — a practice that left many two-digit date fields intact.

As a result, the year 2000 will appear to many systems as 00, not 2000, and will be interpreted as 1900. And lately it seems as if an entire industry has arisen to gauge the implications of this misinterpretation.

Trade-offs from the '60s lead to strategic decisions today

The push by those early programmers for memory economy has today's executives facing critical, strategic infrastructure decisions.

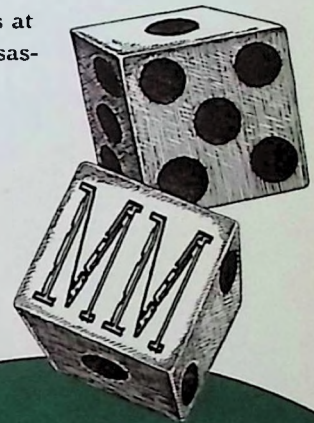
Because the implications of wrong dates on a universal scale are not yet truly known, enterprise IS management and supplier management cannot even fully describe the extent of the problem to their peers, much less provide comprehensive solutions. Information systems today are so widely integrated that finding every likely corrupt transaction resulting from bad-date calculations is an immense challenge. Not finding and correcting bad date fields will disrupt business. And in some cases, those disruptions will be severe.

Some analysts look at the Year 2000 problem and say the sky is falling. Others say the problem will be less cataclysmic than originally thought. Either way, the Aberdeen Group views the Year 2000 problem as much

about competitive advantage as it is about technology. In fact, even executives whose roles have nothing to do with technology will find themselves playing pivotal roles in finding Year 2000 solutions for their enterprises.

The key is to start searching for those solutions right away. Many organizations are waiting for a "magic pill" technology to miraculously solve the Year 2000 date problem. This approach is at best a craps shoot, with potentially disastrous implications.

► This is Part 1 of a 2-part Solutions Series on the Year 2000. Look for Part 2, which will discuss specific solutions, in the June 2nd Issue of Computerworld.



ILLUSTRATIONS BY TIM BRINTON

The lead article in this Solutions Series was co-written by Peter S. Kastner and Susan L. Irving of Aberdeen Group Inc., a market research and consulting firm in Boston. As Group Vice President of Aberdeen, and general manager of its commercial systems practice, Peter Kastner analyzes trends in databases, OLTP, decision support, client/server architectures and distributed commercial systems development. He also conducts studies in distributed systems planning, electronic commerce and database management software issues for suppliers and user organizations. Susan Irving is senior analyst for Professional Services Research. By focusing on the economics of the business from the end-user's perspective, she assists senior executives in analyzing the benefits of using professional services to achieve their enterprise objectives.

The

By Peter Kastner
and Susan Irving
Aberdeen Group

Year 2000 problem will rock businesses with the same force as any sudden, unplanned for market change.

Look at how the competitive landscape of the airline industry has shifted abruptly several times in the past 20 years due to regulatory changes, the use of computerized reservation systems and scheduling schemes, and international alliances. The catalyst for each shift differed but the out-

prises by ensuring Year 2000 safety. For this latter group, the problem will be the catalyst for change that will in turn create opportunity.

Not just about software

The Year 2000 problem is not necessarily about software. For a company making consumer packaged goods, the problem is about the gain — or loss — of shelf space. (No superstore chain will let a vendor near its stores without proof of Year 2000 readiness for any integrated inventory and provisioning systems.) A process company may find that its Year 2000 problems are about its ability to efficiently and competitively use its asset base. A services company may find that

Year 2000 bugs undermine its ability to mine data marts for targeted-customer marketing.

For all businesses, the unifying theme is this: In the best case, the Year 2000 problem is about determining competitive advantage; in the worst case, corporate viability.

Why competitive advantage? Tightly integrated, Web-enabled technology infrastructures are becoming the circulatory system of companies. Global organizations are interdependent on realtime, online worldwide information that can be transmitted from Singapore to Atlanta with a single keystroke.

In many cases, these enterprises are also technologically intertwined with the operations of their customers, suppliers and distributors. If this information flow is compromised, so is their business. And if there is one thing the Year 2000 bug promises, it is to compromise critical information flows within and between enterprises.

How Enterprise Executives Can Play a Pivotal Role in the Year 2000 Solution

Companies that are not
Year 2000 safe
will see their competitive
strengths eroded.

come was the same: a reshuffling of industry players. Inevitably, the winners were the companies who found opportunity — not peril — in change.

In the same way, the Year 2000 problem will bring market disruption — perhaps upheaval — to technology-enabled enterprises and industries. The task of hammering out the competitive strategy and plan for survival will fall on the shoulders of senior executives. Some of these

executives will delay making decisions and find themselves in a reactive mode as they scramble for resources and technology to ensure the continued operation of their businesses. Others will attempt to strengthen the competitive footing of their enter-

I have to
reach my
Year 2000
deadline.

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Ardes 2k incorporates a tools-neutral process bound only by the best automation software available. The key is a unique Modular Repeatable Process (MRP) that can be customized for individual environments, dynamically scaled to any size organization, and configured to measure any quantifiable results. As a result, your cost and training time is effectively minimized, while your enterprise's productivity and efficiency is significantly increased.

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Data Dimensions' unique technological process lets you easily transfer Ardes 2k knowledge to your information systems personnel and effectively implement solutions. By utilizing CD-ROM and Internet technology, Ardes 2k Year 2000 solutions are now available to a worldwide client base.

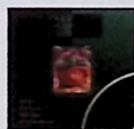
Comprehensive Enterprise-Wide Solutions

Ardes 2k is the most comprehensive millennium update package available. The easy-to-use format provides technical information that can be optimized for your enterprise environment via CD-ROM, Internet Web site access, complementary support tools, products and services.

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Ardes 2k is the only proven solution that can help your organization meet the complex challenge of 21st century compliance.

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The same interdependence can be found in small and mid-sized companies who use technology to secure relationships with larger companies, to market an amplified presence to targeted customers or to perform vital customer-service functions. No company will be unaffected — the difference will be the degree to which a particular company's

in their area of responsibility and ensure that their 1997 plans address these issues thoroughly.

► CFOs need to be confident that they have all the information to make informed funding decisions for the next five years. They also need to know the potential impact on financial statements and, if their company

nature aware of this problem), must have a persuasive business case for Year 2000 funding outside an IT budget. If this is not possible, they must be prepared to offer results-based business cases for saving one mission-critical IT initiative while shelving another to free up resources for the Year 2000 transition. Prepare for trade-offs.

► The Board of Directors must have input if the funds to complete this project will have an adverse affect on three to five years of financial projections. They may also have a fiduciary responsibility to ensure that their company is Year 2000 ready for business. The lawyers are certain to sort this one out.

► Most important, the corporate leader — the president and/or CEO — will play a lead role in ensuring that this is not only an organizational priority, but that the enterprise pulls together in a war-time-like effort to meet the non-negotiable deadline of Jan. 1, 2000. This will involve early coordination with human resources, change management and training.

This list highlights only some of the areas that an enterprise's executive team must focus on today if they are to find

CIOs must have a persuasive business case for Year 2000 funding outside an IT budget.

operations are altered.

To minimize the risk of your business being compromised, you must start your Year 2000 transition now. Not being Year 2000 safe by the start of the millennium will erode the competitive strengths of companies who, in turn, will lose to industry players who better manage this transition.

Executives in the spotlight

Executives can play pivotal roles in ensuring Year 2000 safety for their enterprises. But to do so, they will need to do three things:

- Require that a comprehensive and proactive plan is in place;
- Secure trusted external assistance;
- Insist that day-to-day business operations experience minimal disruption.

Above all, executives must marshal the collective Year 2000 efforts with a sense of real urgency. This urgency will send a signal to the rest of the organization that becoming Year 2000 safe is a corporate priority, and that meeting the millennium deadline is non-negotiable.

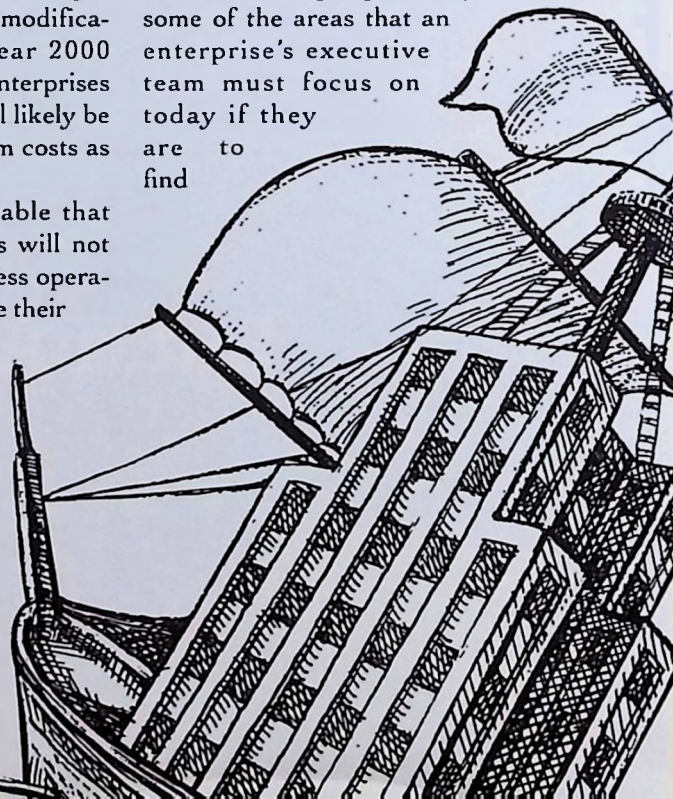
Because Year 2000 is a strategic business problem, rather than a technology problem, each member of a senior executive team needs to identify critical issues resulting from the Year 2000 bug

is publicly held, be prepared with a communication strategy to shareholders and the market at large for 1997 reporting.

► Controllers must check with their audit firm for critical input on how regulatory accounting rulings — for example, the Financial Accounting Standards Board (FASB) in the U.S. — will allow companies to treat these extraordinary costs. FASB is likely to insist that publicly funded companies expense the costs of modifications and repairs for Year 2000 corrections. In contrast, enterprises that replace technology will likely be able to treat the new system costs as a capital expense.

► COOs must feel comfortable that Year 2000 transition plans will not interrupt day-to-day business operations. They must also ensure their involvement in each critical decision. COOs will also have an early vantage point to see problems arise and should have continued participation on the Year 2000 transition team to forestall these problems.

► CIOs and Chief Technology Officers, or CTOs (two positions that are by



Year 2000 Best Practice Recommendations

This list of executive best practices, culled by Aberdeen Group from Year 2000 transition efforts, service providers and top Web sites, will grow – and change in emphasis – as we near Year 2000 transition triggers.

- 1. Start immediately!** The Year 2000 problem is large, complex and, as yet, undefined. The only given is that it will get bigger for enterprises that take a wait-and-see approach. The cost of external support will skyrocket as resources are claimed and quality will suffer as companies rush towards a solution. An early start will give enterprises time to deploy solutions strategically, rather than reactively.
- 2. Stop wounding Year 2000 messengers.** Word in the IT cubicles is that there are lots of wounded Year 2000 messengers. Don't be surprised if information about the real magnitude of resource requirements for this project has not yet reached executives in your company.
- 3. Send lawyers, guns and money.** (Substitute "accountants" for "guns" and your company has the start of the resources required for a Year 2000 effort). Issues need to be understood throughout the organization. Accountants can provide details on current regulatory thinking on how to treat project costs — answers that may impact an (expensed) repair vs. a (capitalized) replace decision. Lawyers will educate themselves on an enterprise's potential exposure from the transfer of corrupt data or the inability to deliver product due to a technology error.
- 4. Request funding for the whole project upfront.** Using best estimates and all the trusted external assistance they can secure, CIOs should develop five-year project estimates (assuming two years of post-2000 conversion work) for unique Year 2000 transition requirements. This will be a thankless task as you present this to the board. But laying out the magnitude of the task ahead will allow other critical enterprise resource decisions and planning to be accomplished. Estimating total costs will also strengthen your case for management, rather than IT, funding later on.
- 5. Treat this as a war-time effort.** Fostering a sense of urgency and common purpose throughout your organization will lead to a more focused and comprehensive effort to meet the non-negotiable deadline. This is not just an IT conversion; it is a matter of competitive viability.
- 6. Secure trusted outside resources quickly.** Many vendors are vying for your Year 2000 transition business. In 12 months, this will be a seller's market. When assessing vendors, ask them what they will not be able to do for your business. The best service providers will tell you this.
- 7. Check hardware, systems software and networking compliance.** Conventional fixes treat Year 2000 problems only in applications. There are date-field problems in networking and hardware equipment and systems software that often times are not inventoried — and, ultimately, not fixed. Executives need to feel comfortable that their project team is addressing technologies outside of applications. Executives should insist on vendor documentation and certification of the Year 2000-readiness of their equipment. This compliance documentation should also be secured for all purchased applications.
- 8. Insist on seeing external partner compliance plans, updates and completion milestones.** And be prepared to make your own plans available to partners. Ironically, the enterprises that are the most advanced in linking their operations to the operations of their vendors, customers and distributors are most at risk for corrupt data. Executives need to ensure that all external links are Year 2000 safe — and that they can provide this safety back.
- 9. Put the community of Year 2000 resources to work.** Encourage the project team to use (if they don't already) the Internet communities that have sprung up to tackle the problem. In addition, regional user groups offer a lot of valuable shared experiential advice.
- 10. Beware the Law of Unintended Consequences.** Getting to Year 2000 compliance will likely require new software versions of compilers, operating systems, utilities and other infrastructure software. Expect problems induced from the infrastructure changes before you even change one line of Year 2000 source code. Also, plan for performance tuning as part of the test cycle.
- 11. Staff a post-2000 SWAT team.** There will be lots of work to do after 1/1/2000. Even the best efforts may not trap and fix all dates. And some disruptions from bad dates will not be discovered for months. There will also be unanticipated sources of external data that could corrupt your information. A team should be rested, ready and eager to tackle whatever comes up as the century changes.
- 12. Save the blame for January 1, 2000.** Many companies are currently engaged in time-wasting discussions aimed at affixing blame for the problem. Companies should not blame anyone — yet. Casting blame today diverts attention away from the real need to fix the date problem quickly, effectively and comprehensively. On Jan. 1, 2000, the finger will most likely be pointing at vintage 1997 decision-makers who refused to acknowledge this problem and who did not allocate the proper resources and time early enough to fix the problem.
- 13. Don't use misfits and re-treads.** Give the troops fighting the Year 2000 war a rosy picture of their post-war career path.

Year 2000 Issues: Understanding Will Lead to Action

By John R. Logan, President, Aberdeen Group, Inc.

Perhaps one of the most startling aspects about the Year 2000 problem is how few enterprises have a comprehensive, realistic plan to deal with it. Yes, technical repair plans are in place — but few IS professionals believe that these can be accomplished in time, within budget and without disrupting business operations.

Even those enterprises that are well on the way to fixing their legacy applications are pulling staff off high-priority application development and implementation projects, because the assessment phase of their Year 2000 repair efforts has started to uncover more problem areas than originally anticipated.

As these new technical problems surface, even the most competent senior enterprise executives are not prepared to proactively manage the Year 2000 repair process. No organization confronting a crisis can long afford to tolerate this type of situation.

Executives, both in IS and at senior operations levels, still do not feel they thoroughly understand the issues involved in managing through the Year 2000 problem.

Even more unsettling is the fact that, as their knowledge increases, all they see are higher expenses that are not covered by today's budgets. In the private sector, this formula often translates into lower profits; in the public sector, it translates into budget overruns that must be explained to skeptical oversight boards.

Look at the long term

To manage through the Year 2000 transition, you must realize that it is a long-term extended enterprise information infrastructure and operations business issue. Once the true scope of the problem is understood, there are four actions that should be part of every plan:

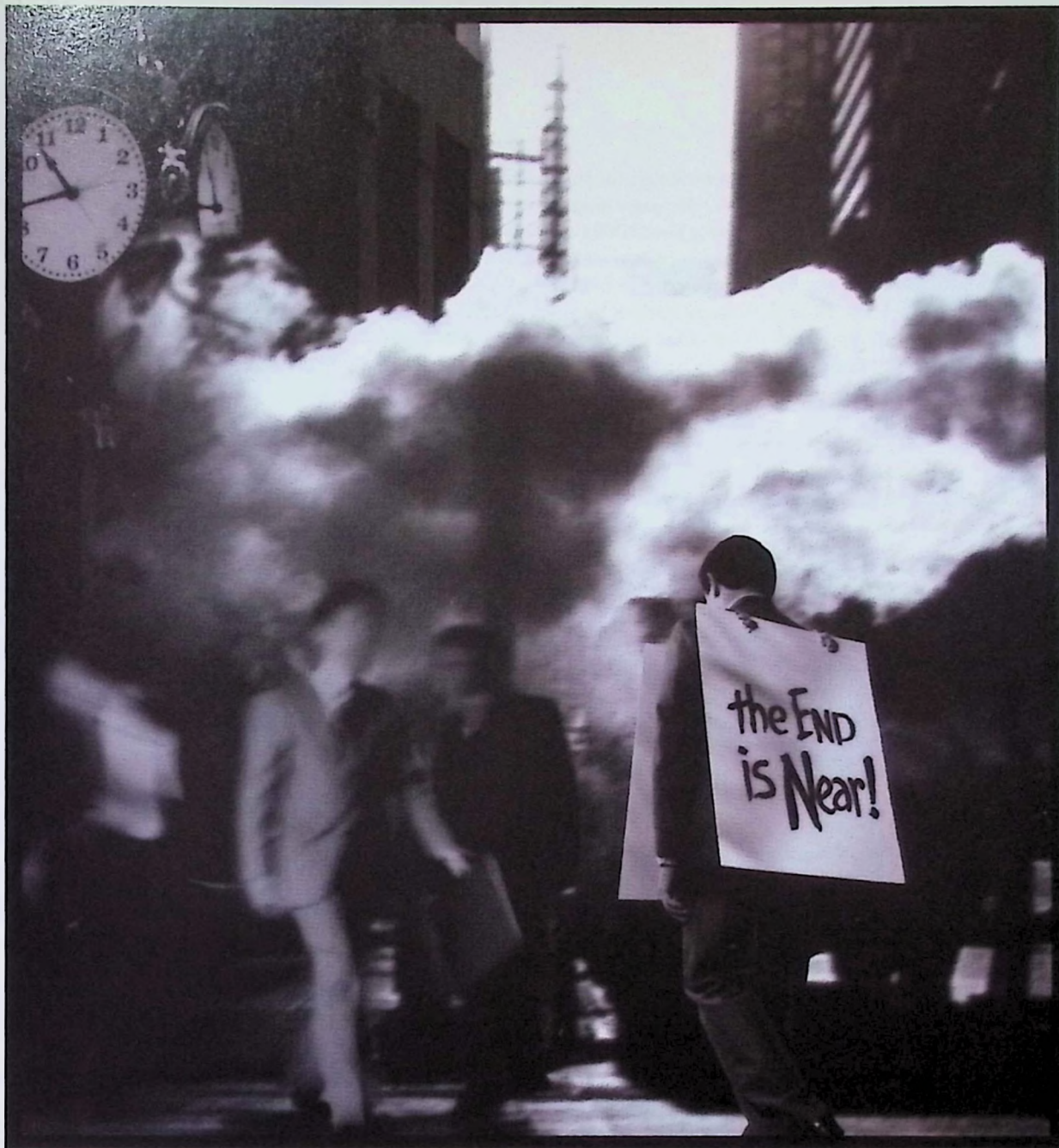
⌚ Where possible, replace all technology components that cannot effectively support Year 2000 safe date codes. (Replacement may appear more costly than repair — until risk factors, lost opportunity costs and financial considerations are taken into account.) Appoint a Year 2000 task force to make implementation decisions right away. (And yes, this will be a high-stress position. The impact of many Year 2000-related decisions are difficult, if not impossible, to quantify.)

⌚ For non-compliant applications and components that must be repaired because they cannot be replaced, determine the lowest level of functionality your enterprise must have from them. Put these "good-enough" repaired applications in place as quickly as possible (but by November 1998 at the latest) to be able to take the additional corrective actions required to keep your business operative.

⌚ Plan to test, re-test and re-test again until your entire information infrastructure is proven safe. Be sure to test date-format interfaces with business partners and government agencies with whom you electronically exchange data.

⌚ Work only with hardware, software and professional service suppliers who have a long-term commitment to your success. Managing through Year 2000 issues — technical, business and legal — will be a tough battle. You will need allies and trusted advisors upon whom you can depend.

Every Year 2000 management action plan must project the timing and costs required to establish the desired levels of functionality and quality for your enterprise's information infrastructure today and through the millennium transition. Quite a challenge — and one for which you will need to employ your very best management skills.



Time to call HP.

ONLY 1,006 DAYS LEFT For the information technology executives who hear nothing but doom and gloom about the coming of the year 2000, we offer a brighter picture: one that includes an opportunity to re-think, perhaps re-engineer, and most definitely provide your enterprise with more flexibility and power than ever before. Since you need to fix the Year 2000 problem anyway, we suggest you do it using

the latest Internet-enabled, client-server technology that prepares for the future instead of echoing the past. With our world-class partners, we can help provide everything you need, from decision support and data warehousing applications to mission critical environments. The bottom line: if you haven't contacted us already, it's time to do it now. Capitalize on chaos. Call 1-800-HP-KNOWS.

*Solving the Year
2000 problem
involves high-level
planning and
accelerated execution.*

Year 2000 solutions.

Many others exist. For instance, who will be responsible for ensuring that any technologically linked vendors, customers, suppliers or distributors are also Year 2000 safe and will not be able to corrupt the information in your systems? And as Year 2000 problems become operationally apparent and business-changing decisions need to be made, who will be responsible for corporate-wide employee communications?

This list also does not account for the equally critical work of the project team, including finding and using tools to correct bad dates, developing and verifying lists of Year 2000-compliant hardware, software and networking vendors (both for existing systems and replacement options), and managing staff morale on a project where the nature of the work is often monotonous and career-limiting.

Obtain outside help

Most companies, lacking the resources, time or comprehensive knowledge to address the breadth of Year 2000 problems, will require external assistance for their Year 2000 transitions. Yet there are no one-stop shopping options for such solutions;

today's providers are focusing on application conversion issues, while overlooking hardware, net-

working and business issues.

Because there are so many pieces to the Year 2000 puzzle, executives should engage a trusted professional services provider to help them navigate the business and technology decisions.

One attractive option is the consulting arm of your audit partner — or a firm referred to you by them. This firm has a stake in the ongoing viability of your business, understands the inner workings of your operations, and is positioned to continue to apply the best practice learning from its collective Year 2000 projects. Although this firm will not provide one-stop shopping services, it will be in a position to understand where to secure additional resources.

If your audit firm's consulting practice is unavailable, there are many other top-notch consulting and conversion firms who can provide support. A list of such vendors can be found at the Web site www.year2000.com. Insist on references from consulting companies with applicable experiences to ensure that you have not merely found an opportunist in a three-year market.

Whomever you choose for external assistance, you need to secure that relationship ASAP. By mid-1997, there will be little capacity left in the industry. This will drive conversion prices up and quality down, and will jeopardize the chances of many companies for a safe Year 2000 transition.

Conclusion

Aberdeen Group believes that there will be enterprises who gain from their Year 2000 solutions by ensuring that their technology infrastructures continue to enable, rather than disable, their business objectives through the millennium transition.

And because no company is an island, these internal technology infrastructures often encompass external partners — vendors, customers and distributors — who must also be able to share data that is not corrupt.

Enterprise executives can play a pivotal role in the Year 2000 solution by bringing to the surface critical business issues that result from Year 2000 repair strategies — and developing and executing the proactive plans needed to deal with these issues.

The upshot is that solving the Year 2000 problem involves high-level planning and accelerated execution. The challenge today is for executives to devise a realistic solution for their enterprise (and their enterprise's partners), secure adequate resources to support this effort, find trusted external consultants, vendors and tools before these resources are claimed, and ensure an active forum for the escalation of critical business decisions that must be decided by senior management.

Executives that meet Year 2000 challenges head-on will lead their enterprises through a more graceful transition into the next millennium while gaining competitive ground on rivals who are not Year 2000 ready. ☞

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This Solutions Series on the Year 2000 issue was created by Computerworld Custom Publications. For reprints, contact Heidi Broadley at (508) 820-8536 or at heidi_broadley@cw.com.

The Aberdeen Group is at (617) 723-7890. Peter Kastner can be contacted by E-mail at kastner@aberdeen.com.

Questions or comments on the second story in this supplement should be forwarded to managing editor Peter Bochner at (508) 820-8289 or peter_bochner@cw.com.

Part 2 of this Series will deal with specific solutions to handle the various pieces of the Year 2000 puzzle. It will appear in the June 2nd issue of Computerworld. Editorial inquiries should be addressed to Peter Bochner.



By Anna Tortig

a lousy job, but somebody's got to do it . . . managing year 2000 date-change projects, that is.

The Year 2000 problem presents IT professionals with immense project management and organizational challenges. First they have to convince CEOs and other top managers that they should fork over \$1 million or more to rectify a problem that was in fact created by data pro-

cessing departments. Then they have to convince programmers to put the brakes on career advancement and take on the boring, tedious and time-consuming task of checking thousands and thousands of lines of computer code for any potential failures when the clock strikes 12 on Jan. 1, 2000.

There's neither visible return on investment for management nor glory for the troops in the trenches. But there are penalties galore, particularly for CIOs. Thomas D. Oleson, an analyst at International Data Corp. in Framingham, Mass., puts it bluntly: "For CIOs, this is a career-ending project if they screw it up."

And screw-ups could have dire repercussions for an organization when it comes to Year 2000. "This is a stay-in-business situation," says Wayne Johnson, senior project manager at Reliastar Financial Corp., a life insurance and annuities company in Minneapolis that in the next 21 months will pump \$39 million into changing 15 million to 20 million lines of code

across all its applications. Notes Johnson: "It touches just about any line of code we have."

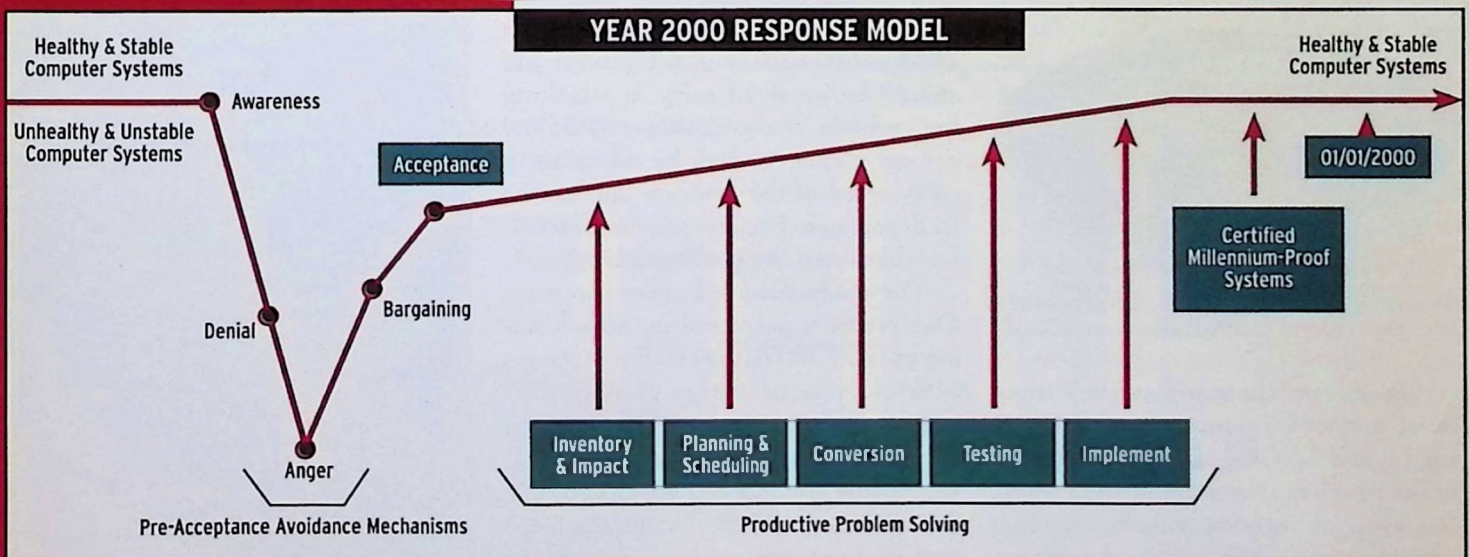
Because it started early, Reliastar should be safe. But IDC's Oleson estimates that 10% of large corporations will not finish the task of conversion by the year 2000 deadline. Others, he notes, will meet the dead-

Management: Key to Year 2000 Success



There are penalties galore, particularly for CIOs.

Others, he notes, will meet the dead-



Source: Kappelman & Cappel, 1996

line but will still be finding bugs after the fateful day. (Perhaps long after. Bugs don't always show up right away. In a program Oleson wrote as an insurance actuary in the 1960s, "the bugs didn't show up until the end of the 1970s," he notes, "because we didn't go down that branch of the program.")

of concern varies by industry. In the insurance field, where dates are in nearly every line of code, 78% of IS managers consider the date-change problem a major concern. In banking, 64% of IS executives feel that way. But in the manufacturing segment, only 41% of IS executives, 20% of CFOs and

charge of the situation.

Those who were in charge reported that they were spending less than a third of their time on the project. Only 16% were devoting more than 90% of their time. That means most Year 2000 managers are not devoting full-time attention to the project, and, according to the study's author, "if these figures are even half true, they portend of a serious degree of very negative year 2000-related consequences."

The SIM report also suggests that important projects, particularly in the area of application development, may

Only 16% of those heading up Year 2000 projects are devoting more than 90% of their time.

What will happen to the laggards — the companies that don't make the deadline? Theresa O'Neil, vice president of marketing, data warehousing, for Platinum Technology, offers one scenario. "They'll return to paper forms, because their computers won't process the right information," she says. "Payroll and counting will get done, but other applications will get put on hold."

10% of CEOs believe that Year 2000 is a major business issue.

CIOs tend to see the problem as a major business issue more often than CEOs or CFOs. (That's understandable, since they're the ones whose careers are on the line.) But there are many other blind spots in the recognition of the problem.

IS leads in awareness

According to a study by the Society for Information Management (SIM), based in Chicago, the perceived level of organizational Year 2000 awareness does not extend much beyond the IS department. The study noted a very low incidence of reported discussions with auditors and legal staff, two groups that should be involved early in attacking the problem. It also said that CIOs and project managers may be reluctant to carry word of the problem outside the IS department because they are uncomfortable about the problem's origins.

The study detected other concerns. One problem pertained to those heading up Year 2000 projects. For instance, 61% of those in charge of Year 2000 projects have had the assignment for six months or less; only 11% have been in the saddle for more than 12 months. Worse, about 35% of organizations polled appeared to have no one in

You're in good hands with . . .

Several insurance companies have begun issuing Year 2000 insurance policies to cover business disruptions that may result when systems are converted to handle the date change.

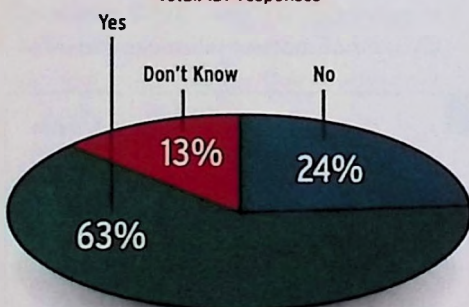
These insurers are expected to be very selective in insuring companies. If a company does not have a comprehensive Year 2000 plan, or if it does have one but does not execute it, it might not be able to collect.

So far users seem less than bowled over by the policy. One attendee at a Year 2000 conference said, "I can just see going up to my boss and saying, 'Our systems don't work. But don't worry, we're covered.'"

But if nothing else, such an insurance policy may be "brilliant in terms of raising management awareness," says Don Estes, a Year 2000 consultant in Lexington, Mass. "Now you can go to the CFO and speak in terms he understands."

Is your company's executive management aware of the real risks involved in this project?

Total: 157 responses



Comment: One in four executive managers are unaware of the real risks of the Y2K project. The question is, who's going to bring them up to speed?

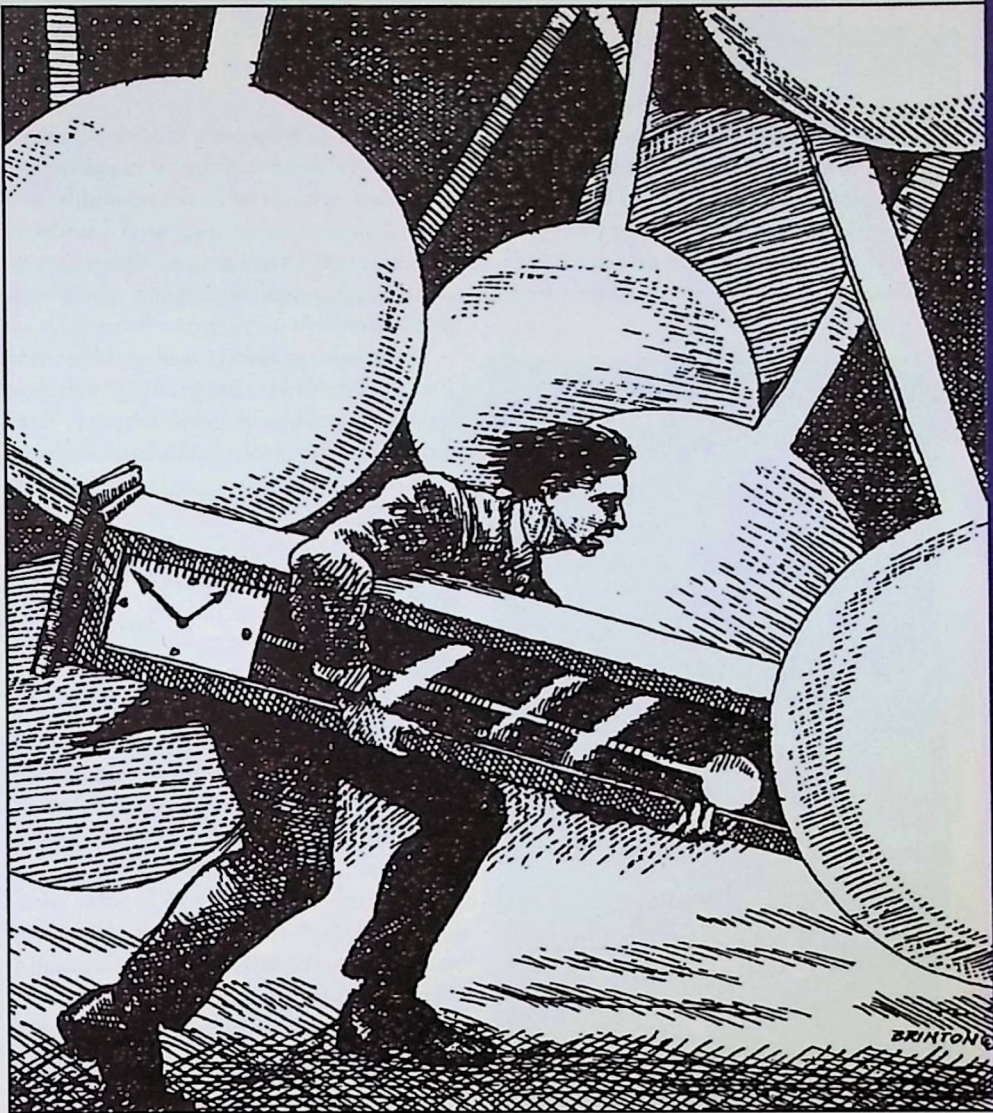
SOURCE: Year 2000 Information Center

Of course, the good news is that most corporate managers — CEOs, COOs and CFOs — are already aware of the problem and understand the consequences of not complying in time. But according to an IDC study, the degree

be ignored as a result of the focus on Year 2000. The flip side may be true as well. Europe, notes O'Neil of Platinum Technology, "is having a heck of a time getting corporate buy-in on the Year 2000 issue. Their focus is on the issue of European currency."

Leon Kappelman, co-leader of the SIM 2000 Working Group and an associate professor at the University of North Texas, says it is crucial to assess the risks — "If it has anything to do with a chip, it is potentially at risk," — and convey that message to management. He said it may be enough to familiarize CEOs with the ramifications of the matter by setting up a test of a mission-critical system and showing what will happen if the company doesn't change its code. In other cases, outside pressure (for example, from an industry group) may be required. The Securities Industry Association has created a Year 2000 committee to ensure that securities companies, banks, stock exchanges and their business partners are compliant. The group plans to conduct a compliance test of Wall Street companies in early 1999.

But for CIOs, getting a message across to top management can be tough. CIOs who are currently involved in Year 2000 projects say that when they brought their boss the bad news, the initial reaction was negative. "The reaction by most CEOs is the same: Why did DP do this in the first place?" says



Roland Laferriere, CIO of BJ's Wholesale Club, a membership-only warehouse club headquartered in Natick, Mass. Laferriere admits to having felt

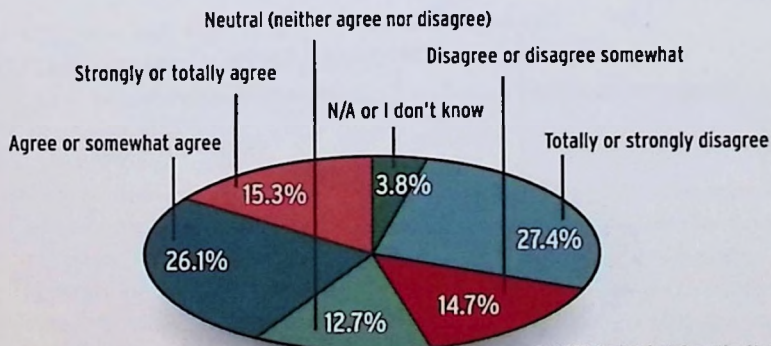
chagrined when he explained the root of the problem to the president of BJ's. "CEOs understand what happened, but still can't comprehend why," he says.

Dave Iacino, the project manager of BankBoston's millennium management team, reports encountering the same anger and denial when he first went to the Board. But those emotions were quickly replaced by the realization that the problem was real. "Because the CTO had done his spadework people knew how to spell millennium," he says. "Funds were committed almost on the spot."

Pressured CIOs can try to turn this tension-ridden situation to their favor through careful project management, starting with their initial efforts to gain

Solving this has the highest priority in my organization

Responses of 156 CIOs and IS managers



SOURCE: Society for Information Management, 1997

the ear of top management. For them, the best strategy may be to anticipate management's questions and explain the date-change problem in a way they can easily understand: an article from a business publication or by illustrating a

Even the feds see the light

Even the federal government, which is wrestling with some legacy systems so old that replacement parts are no longer available, seems to be emphasizing management skills over technology as the key to Year 2000 compliance. "The enormous challenge . . . is not technical; it is, rather, managerial," said Joel Willemssen, director of the information resources management, accounting and information management division of the Government Accounting Office (GAO), in testimony Feb. 24 before the Subcommittee on Government Management, Information and Technology, in the House of Representatives. (GAO is the congressional investigating agency.)

"Whether agencies succeed or fail will be largely influenced by the quality of executive leadership and program management," Willemssen added. "It will be imperative for top agency management — including the agency head and the CIO — to not only be fully aware of the importance of this undertaking, but to communicate this awareness and urgency to all agency personnel in such a way that everyone understands why Year 2000 compliance is so important."

Year 2000 failure on a key application.

"We did a slow, gradual preparation, told them what it will cost us and how it will affect operations and business plans," says Laferriere. "The more you can prepare and anticipate questions, the better."

At sporting goods maker Converse Inc., in North Reading, Mass., talks on how to handle the issue began a year ago. CIO Vincent Cafferelli spent a

identifying fields and testing the programs. They are taking an inventory of all programs and tables and identifying how much of the code needs to be changed. The company also hired three contract Cobol programmers, since programs written in Cobol — the leading language for business applications and programs in banking and insurance — lie at the heart of many millennium problems.

Anything that has anything to do with a chip is potentially at risk.

good deal of time with top management reviewing the history of the problem, why it takes so much time and effort to correct, and what the repercussions might be.

Cafferelli's plan was to get the CEO on board first, then to talk to the CFO and senior executives of departments. Showing management some articles from business publications convinced them it wasn't just a case of a computer glitch, but something that "affects the bottom line," he says.

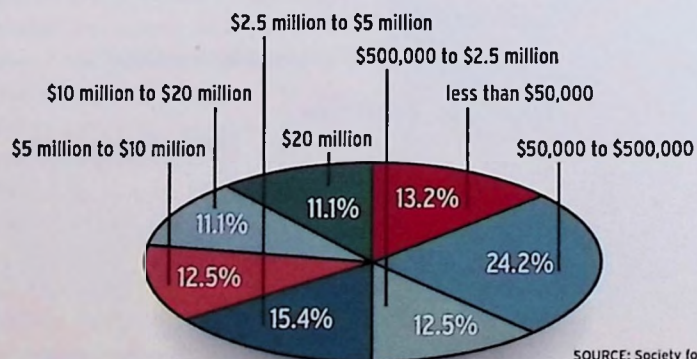
Converse has allocated about 30% of its programming staff primarily to its mainframe systems, going through code,

At Converse, all systems that were used regularly needed some changes. "That amounts to thousands of programs, and 75% of our financial systems," says Cafferelli. Some of the individual programs only required a total of 5% of the code to be changed, others 25% or more.

The issue also forced Converse to decide whether to replace its 10-year-old mainframe, which handles order processing, with a client/server system. "We had to resolve this issue before tackling the Year 2000 problem. The choice was to put in a new system or change this system over the next three

How much do you estimate it will cost to make your organization's information systems Year 2000-compliant?

Responses of 136 CIOs and IS managers



SOURCE: Society for Information Management

years," Cafferelli says. "We decided to change the mainframe because . . . changing to client/server was too risky." Converse aims to complete its Year 2000 conversion by January 1999, giving it 12 months to work the bugs out. Cost to resolve the issue: \$750,000.

For a firm reengineering its business with client/server systems, whose software accommodates four-digit dates, the date-change problem has essentially disappeared. "A lot of our mainframe systems are being moved to client/server, so that aided the Year 2000 part of our reengineering process," says Frank Gladwell, IS manager at Berry Network Inc., a Dayton, Ohio, national marketer of Yellow Pages advertising and a subsidiary of Bell South. Berry and Bell South each have teams working on the date-change problem, which impacts fewer than 5% of Berry's programs, "so we're close to being done," Gladwell says. Berry runs articles in the monthly company newsletter to keep employees up-to-date on its Year 2000 progress.

Killing two birds with one stone

Datapro Information Services Co., a market research firm in Delran, N.J., has already solved its Year 2000 issue with client/server systems, says Martin Levine, director of technology. "With us, it wasn't a Year 2000 issue, but an issue that our business system needed new functions, so we took care of the Year 2000 problem at the same time," he says. "We've reengineered our systems, not only fixing dates but improving the system as well."

The biggest challenge in date conversion is figuring out how one piece of code in one program affects one in another program. Tracking such ripple effects is a seemingly endless task, since testing every branch of every program is impossible, from the standpoints of money and time. To relieve its own programming staff of such tedious work, BJ's decided to outsource mainframe conversion work to ITL-Infosys Tech-

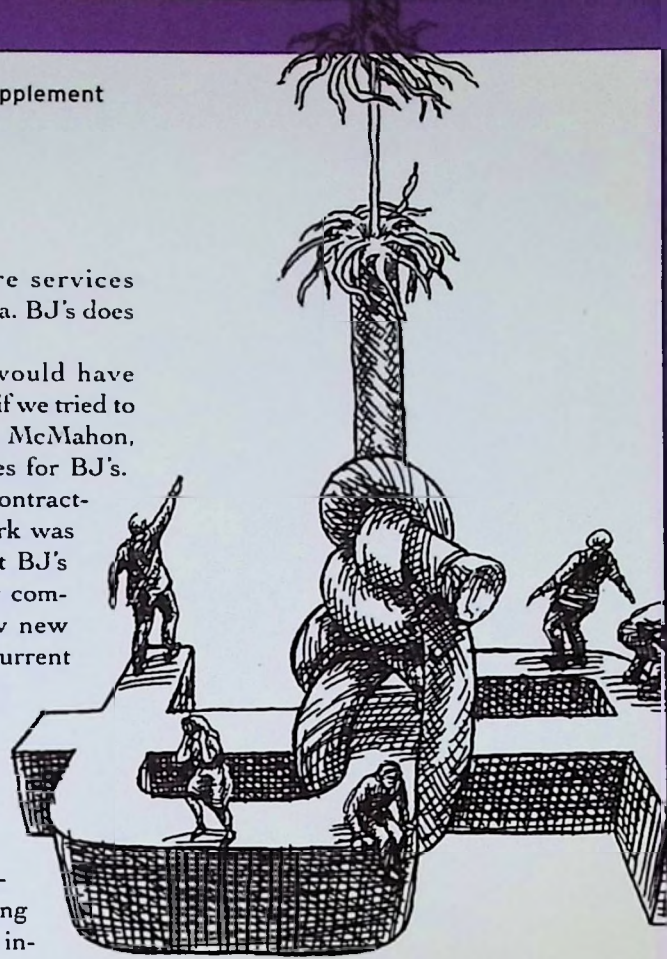
nologies Ltd., a software services provider in Bangalore, India. BJ's does the final testing in-house.

"Our business plans would have been significantly impacted if we tried to do this in-house," says Tom McMahon, manager of systems services for BJ's. IS management felt that contracting out the bulk of the work was the best way to ensure that BJ's would both meet its target compliance date and not slow new development efforts. The current schedule calls for BJ's 10 mainframe applications — which contain 6 million lines of source code — to be converted by July. By then, 80% of its systems will be Year 2000-compliant. BJ's is converting other desktop systems and in-store POS systems itself.

Laferriere figures that BJ's is not only ahead of its competitors in date conversion, but it also has reduced conversion expenses 50-60% over standard U.S. prices of \$1.10 to \$1.50 per line of code. "A critical part of the project was minimizing our staff's exposure to long-term, laborious, boring work that is not career-enhancing," he says.

Cafferelli agrees that a Year 2000

billion distributor of paper supplies completed an analysis showing that less than 5% of its code in affected programs required change. The company is now conducting a pilot project with two tool vendors and an outsourcer. When the project is completed, in the second quarter, "we will develop a plan to see if we can handle this in-house," said project director Bob Niedzwiecki.



How should a CIO who hasn't started Y2K repair efforts go into the CEO's office? With resignation in hand.

solution especially taxes programmers. "Programmers tire easily and get slower as the project progresses. It is extremely tedious work. They must check to see if programs fall through the date field."

United Stationers Supply Co., Inc. of Des Plaines, Ill., is currently deciding whether it needs to outsource part of its Year 2000 work. In February the \$2.3

Niedzwiecki advises CIOs who haven't begun Year 2000 plans to conduct an impact analysis . . . immediately. BJ's CIO Laferriere puts it more bluntly: "If you haven't already started, go in with your resignation in hand." ☞

Anna Tortig is a freelance writer in Cambridge, Mass., who covers technology and related issues.